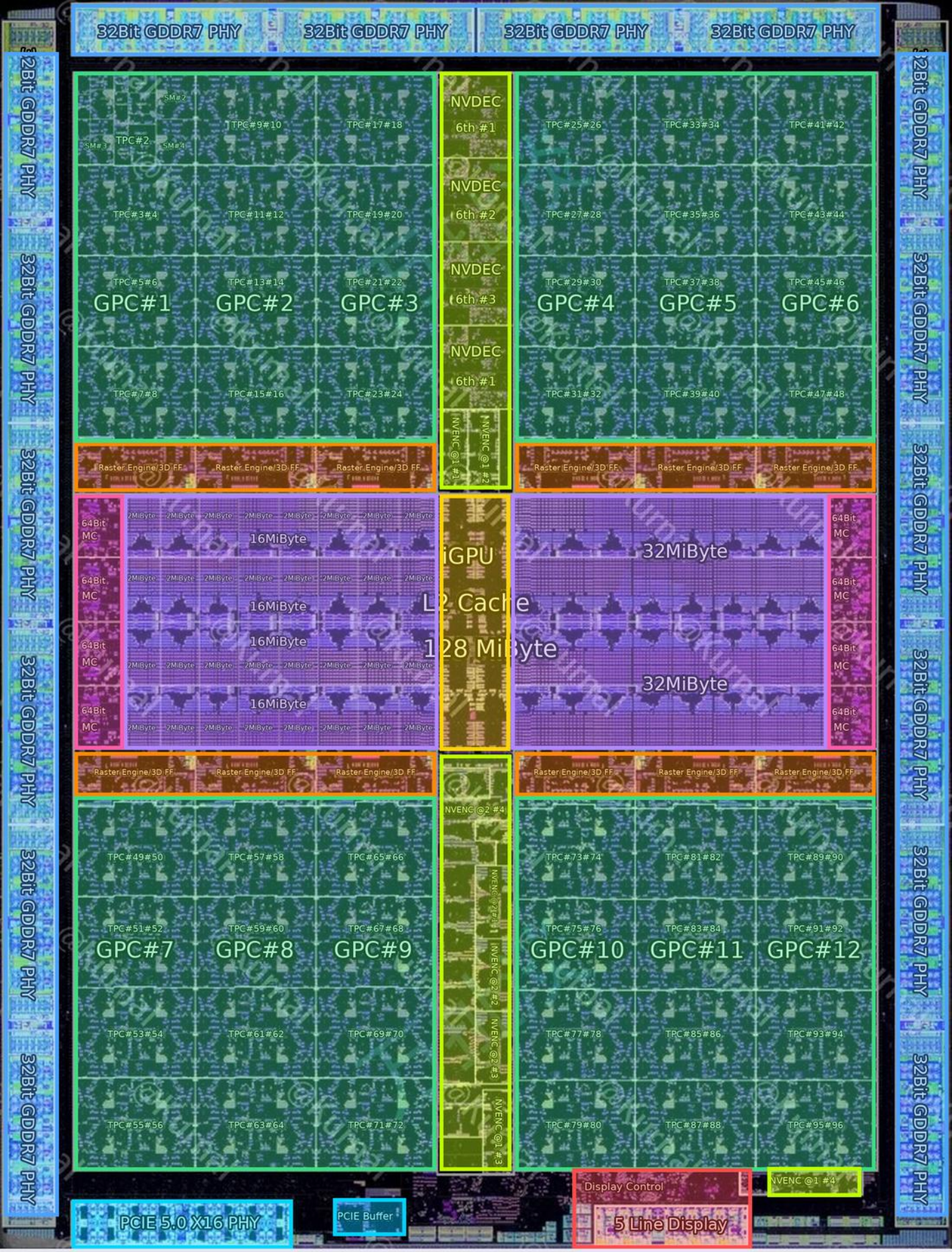
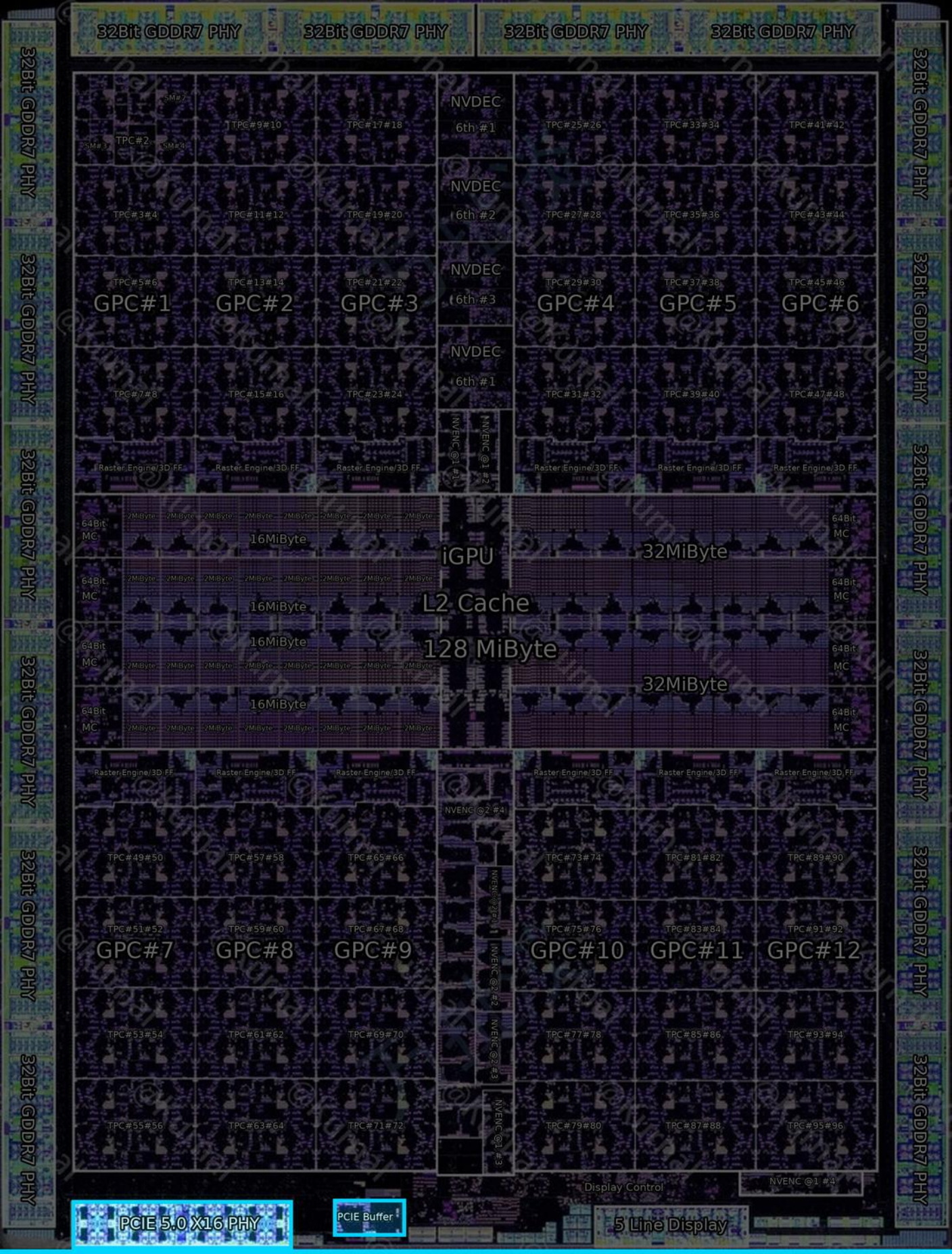


Overview

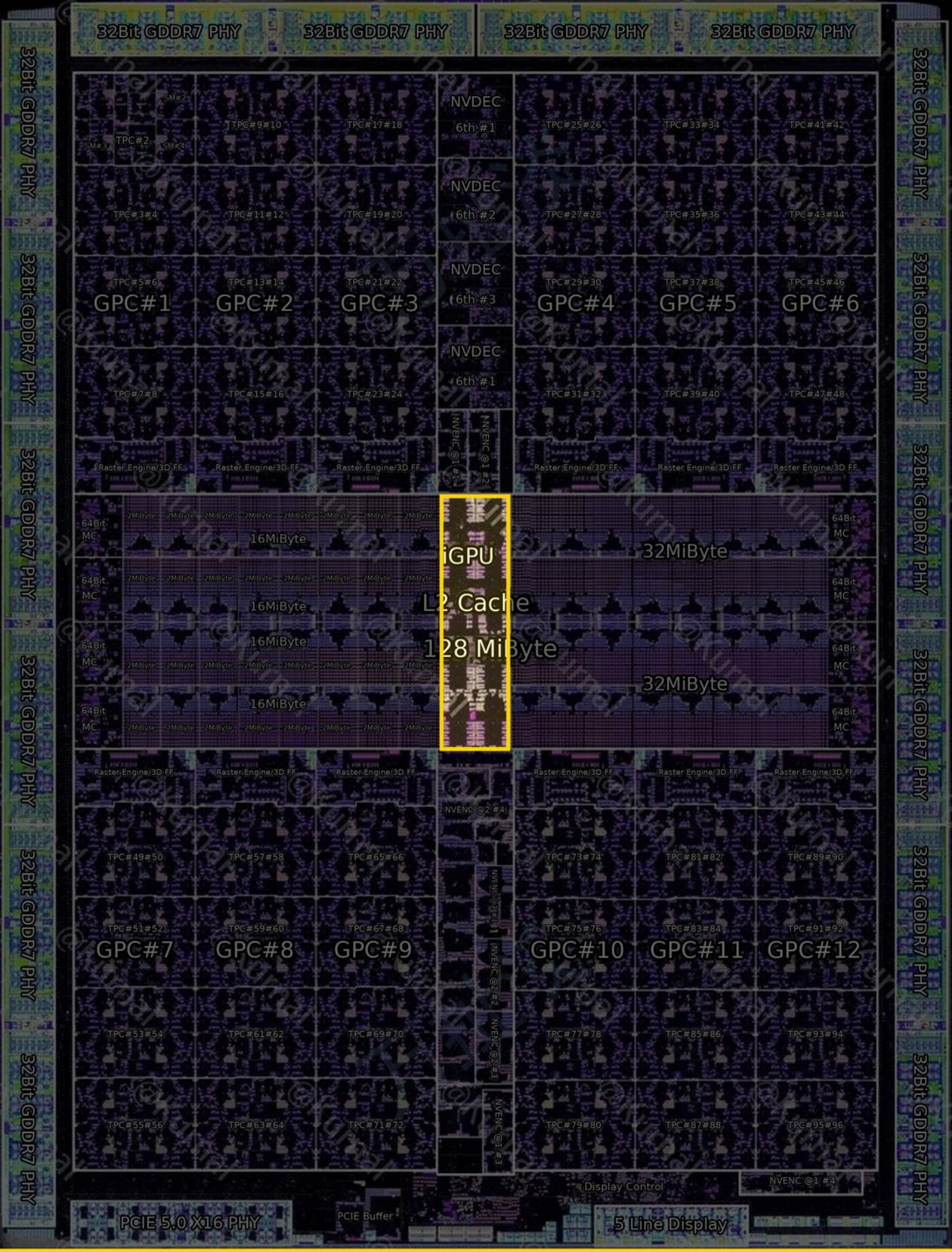
A frame's journey through GB202

Each color is one stop on the path. Real traffic loops (SMs <-> L2 <-> memory many times per frame); this follows one simplified route. Regions come from Kurnal's annotated die shot, and I checked roles against Nvidia's Blackwell whitepaper. Anything marked (guess) is still unconfirmed.



PCIe in

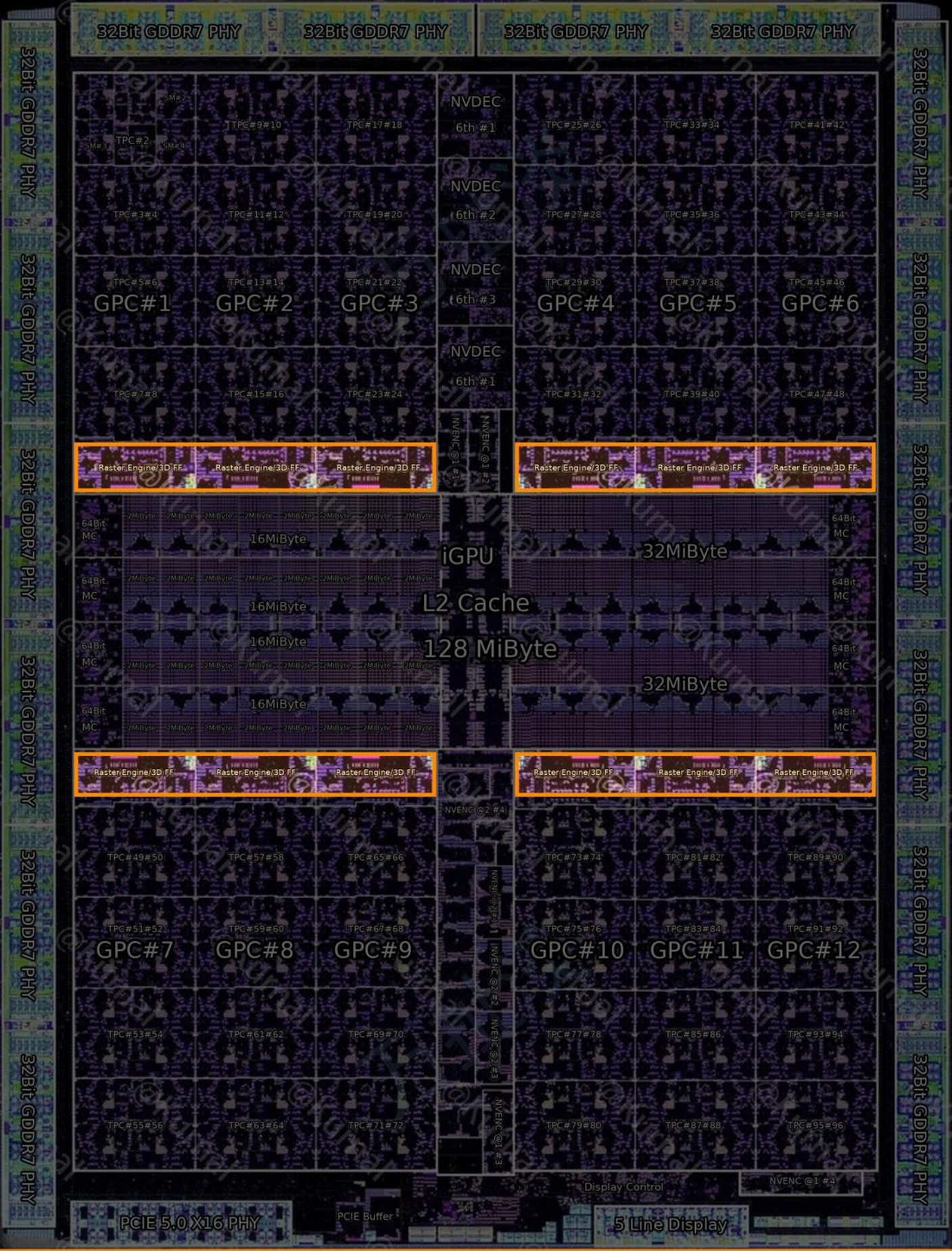
Commands and data from the CPU arrive over PCIe. Nvidia's whitepaper confirms Gen 5; the x16 width comes from the annotation. The PHY is the analog front end. What the buffer does is my guess.



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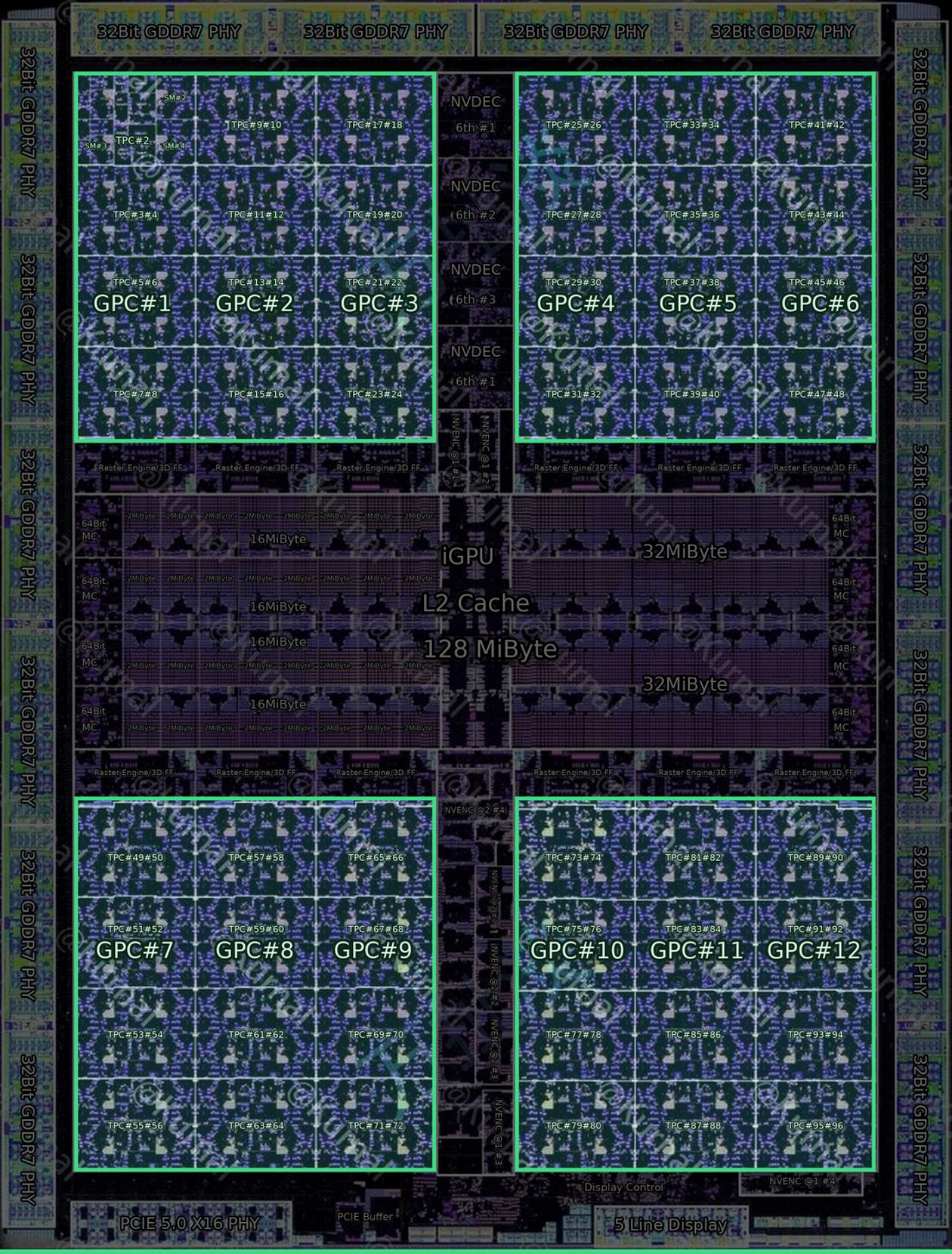
Chip-wide control (guess)

The central strip sits between the two halves of the GPC array. Nvidia describes a front-end processor (the AI Management Processor) that schedules GPU contexts but doesn't say where it sits on the die, so tying it to this strip is my guess. The 'iGPU' label is unexplained.



Raster engines (Raster Engine/3D FF)

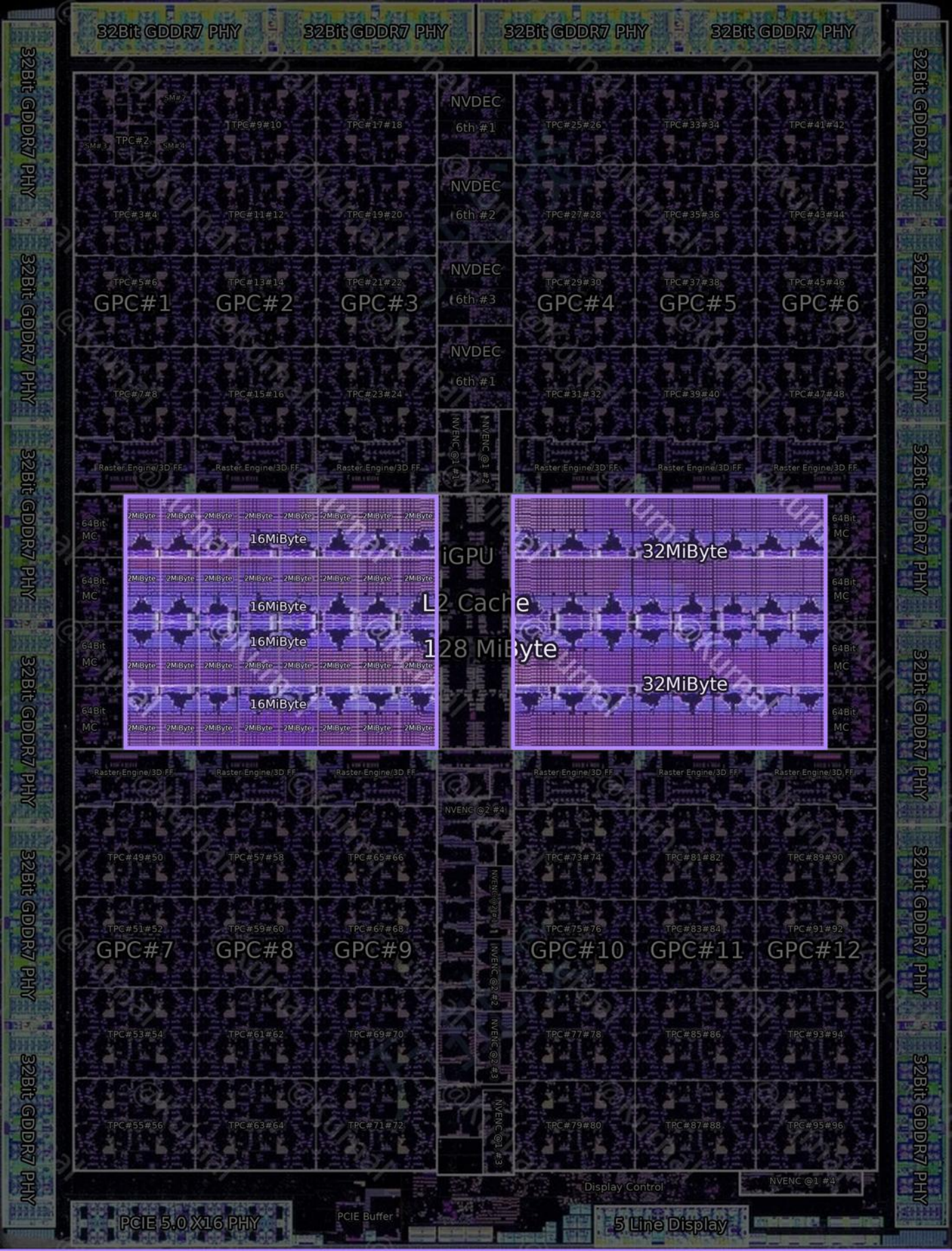
Nvidia says each GPC has a dedicated raster engine and its own ROPs (2 partitions x 8 = 16 per GPC, 192 total). These strips are the raster engines; whether the ROPs are drawn inside them isn't stated. 'FF' = fixed function is my reading.



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Compute: 12 GPCs, 96 TPCs, 192 SMs

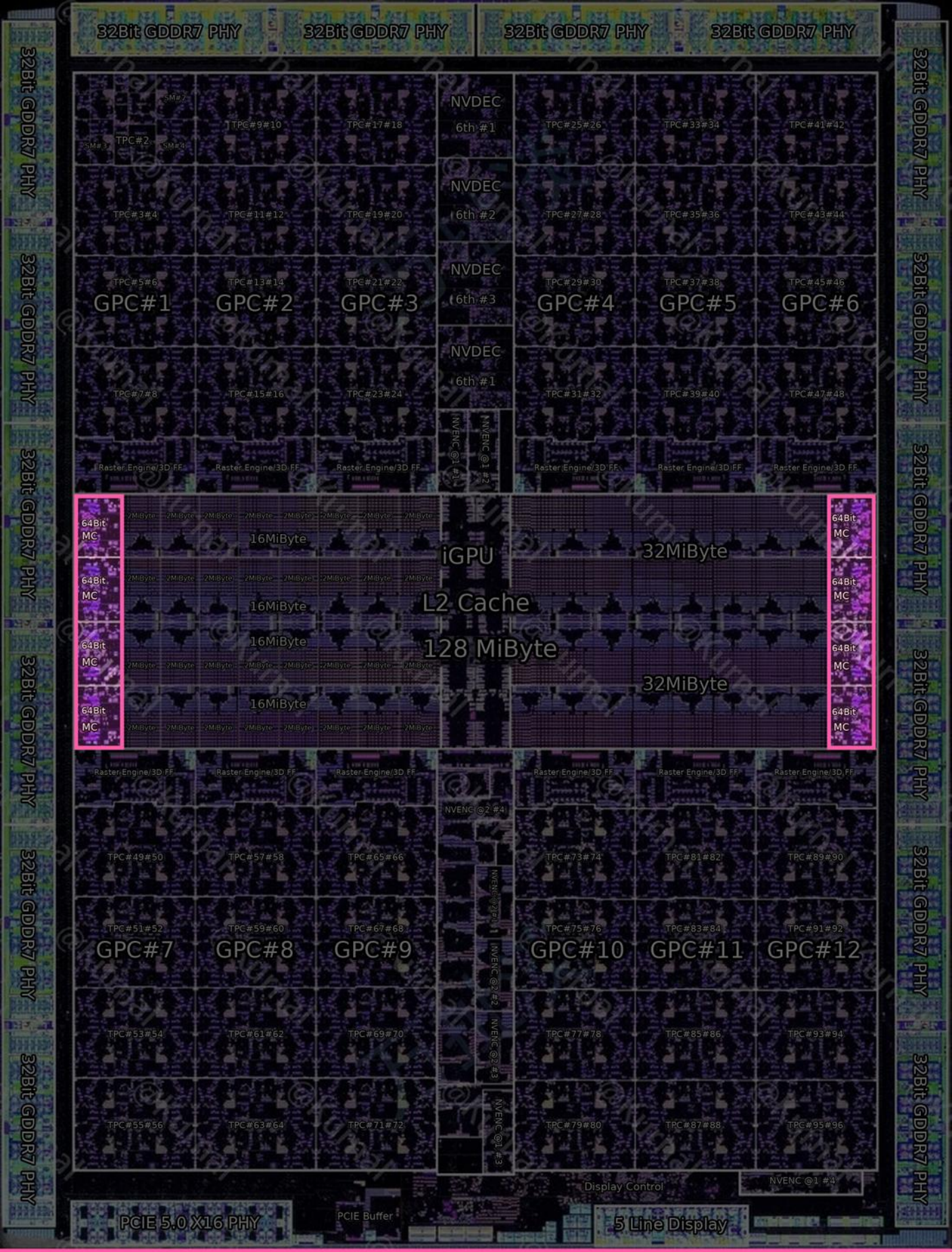
Per Nvidia: 2 SMs per TPC. Each SM has 128 CUDA cores, 4 tensor cores, 1 RT core, 4 texture units, a 256 KB register file and 128 KB L1/shared memory. The RTX 5090 uses 11 GPCs and 170 SMs.



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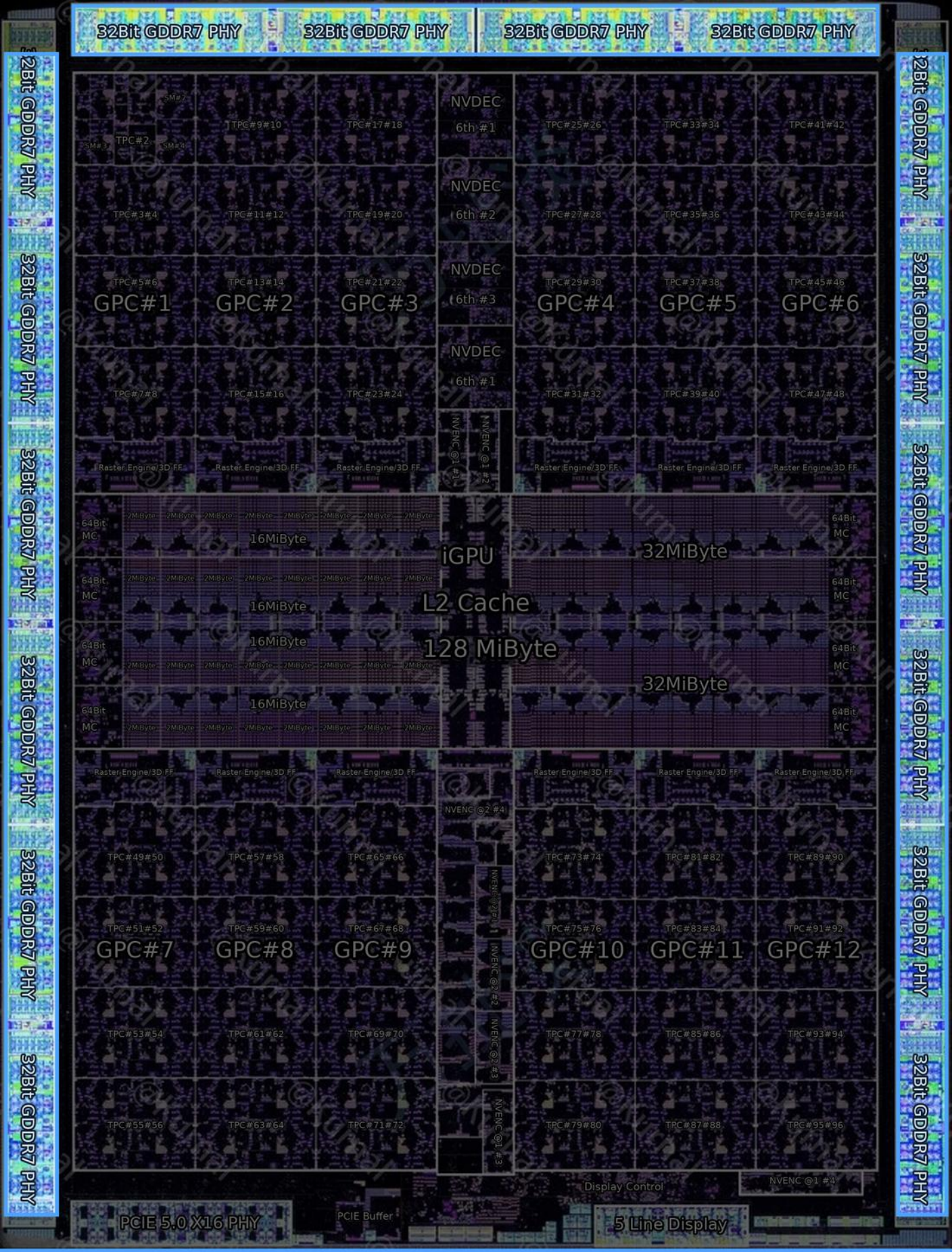
L2 cache: 128 MiB

Shared by every SM. The labels add up (4 x 16 + 2 x 32). The 2 MiB slices imply 64 banks, matching Chips and Cheese counting 64 cache blocks in the die photo (an inference). The 5090 enables 96 MB. A hit here skips a trip to memory.



Memory controllers: 16 x 32-bit

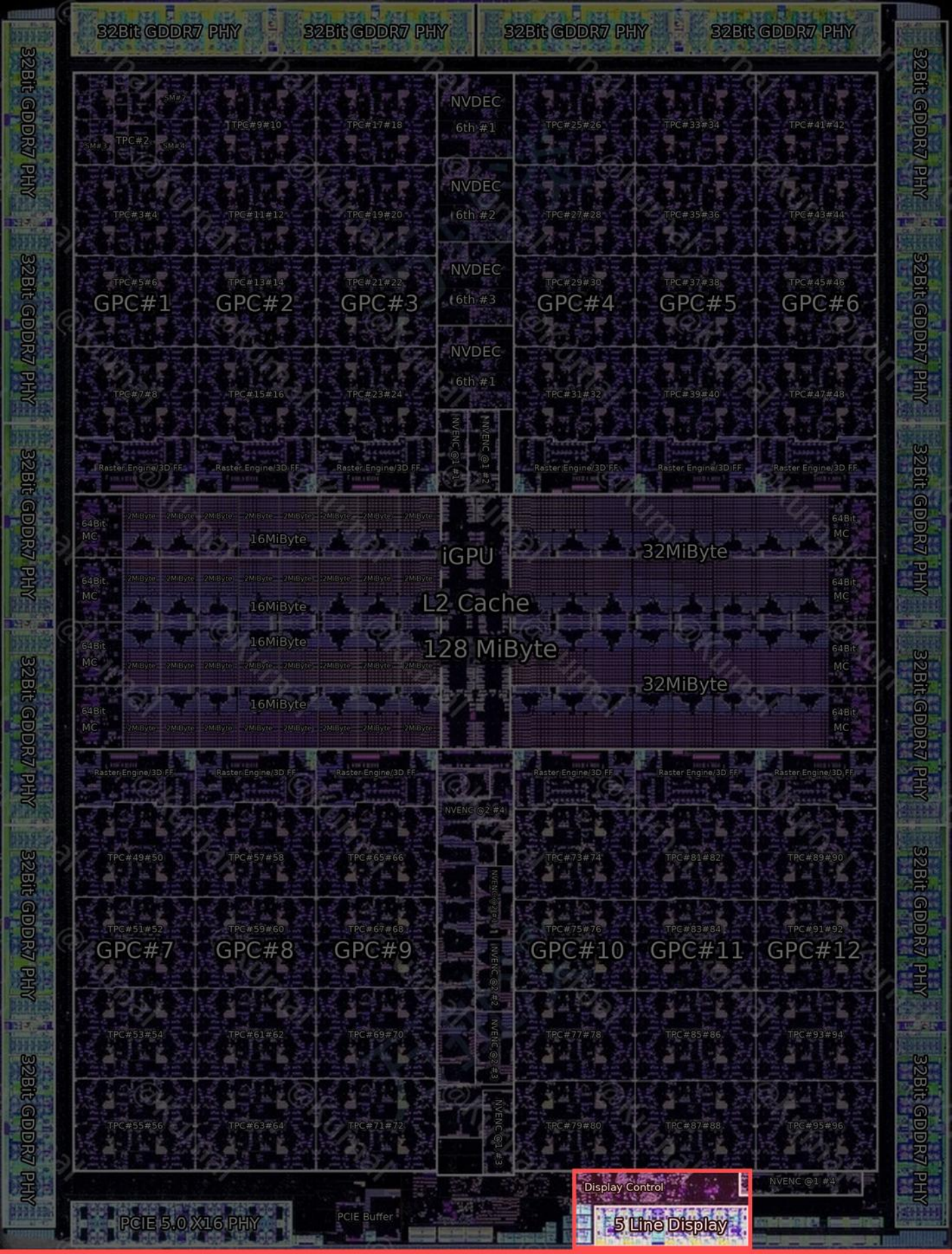
Nvidia describes sixteen 32-bit controllers making the 512-bit bus. The annotator drew them as eight 64-bit blocks, most likely pairs. L2 misses go here to be scheduled as reads and writes.



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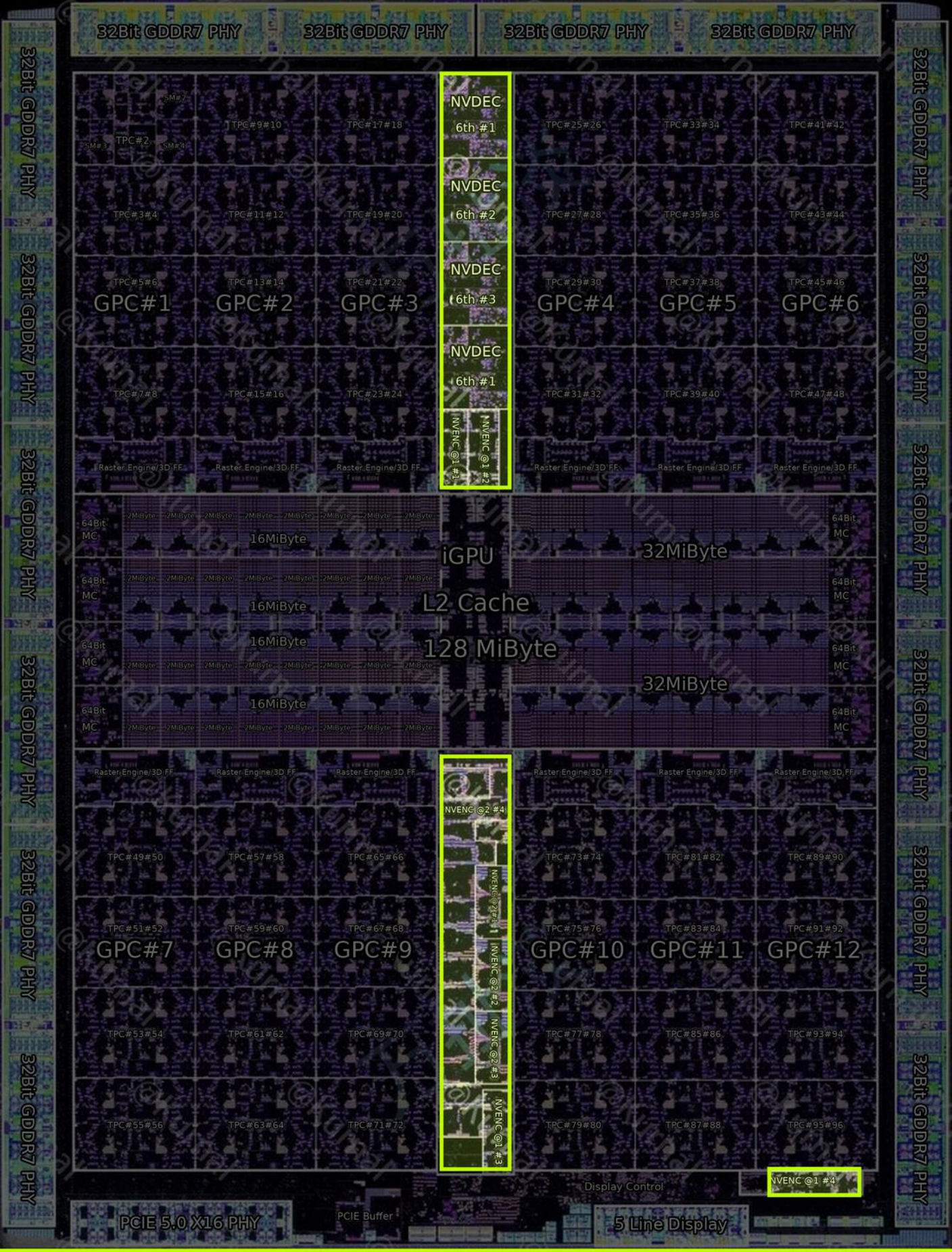
GDDR7 PHYs: 16 x 32-bit

Analog circuits on three edges (4 top, 6 left, 6 right) drive the signals out to the GDDR7 chips, one PHY per chip. The count matches Nvidia's sixteen 32-bit controllers.



Display out

Finished frames are scanned out by the display engine; Nvidia says it supports DisplayPort 2.1b at up to 80 Gbps. The annotation puts display at the bottom edge. What '5 Line Display' refers to is unverified.



Side path

Video engines: NVDEC and NVENC

Fixed hardware for video, separate from the shader cores. Nvidia lists four 9th-gen NVENC and four 6th-gen NVDEC on the full workstation chip; the 5090 enables 3 and 2. The @1/@2 labels are unexplained.